

Are Lexical Representations Separable? Item and Category in Learned Distributed Representations

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Abstract

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1 Introduction

2 Experiment 1: What a Learner Builds from a Known Language

The goal of experiment 1 is to validate our measurement on an artificial language whose structure we control as well as examine how a learner represents that structure.

2.1 Dataset

In order to examine the representation of linguistic concepts, we constructed an artificial language which comprises a probability distribution over a vocabulary of 600 tokens. Every observation is a combination of a **form** (one of 60), a **class** (one of 4), and a **context** (drawn from a pool of 400). The motivation for this setup is to give each word a naturalistic environment: the word occurs in a context that favors a specific class-membership (e.g., *walk* as a verb occurs in a different context than *walk* as a noun). Each form is observed 24 times in each class, every time in a different context, which is what a word in a corpus looks like: one type, many occurrences, each in different company. Consider *walk* used as a verb, in one of those contexts. Four things determine the distribution over the next token. *Walk* itself contributes the same amount whichever class it is used as, and the verb class also contributes the same amount for every verb. The third contribution is comprised of the parts of the representation specific to *walk*-as-a-verb over and above those two: every form and every class has its own random vector, and this term multiplies *walk*'s by the verb's, entry by entry. A sum could be pulled back apart into a form part and a class part; a product cannot, which is what makes this term genuinely joint. The context contributes the

rest, which is noise as far as our question goes, though the model still has to encode it exactly as a language model has to encode the rest of a sentence. The four are weighted, summed and passed through a softmax:

$$P(\text{next token} \mid f, c, x) = \text{softmax} \left(s \left[w_{\alpha} \text{item}(f) + w_{\beta} \text{class}(c) + w_{\gamma} \text{int}(f, c) + w_{\text{ctx}} \text{context}(x) \right] \right) \quad (1)$$

Each term is scaled to unit length before it is weighted, so changing the weights changes the mix of the three effects without changing how much signal there is in total, and the task stays equally hard everywhere in the grid.

The weights w_{α} , w_{β} and w_{γ} in Equation 1 control how much of the true distribution each of the three effects contribute, and we cross them factorially over $\{0, 0.5, 1, 2\}$: zero removes an effect from the language altogether, and the other three values make an effect half as strong as another, equal to it, or twice it. Similarly, w_{ctx} governs how much the context contributes, so we cross those languages with three context weights, ranging from context being a minor part of the true distribution to the dominant one. We train a model on each of the resulting languages at hidden widths of 8, 16, 32, 128, with either a linear or a ReLU activation (Section 2.2). The whole grid is then run with five random seeds, each redrawing the language and reinitializing the model, giving 7,560 runs in all, of which 99.7% trained to a loss plateau.

It is worth noting that entanglement is guaranteed under a specific subset of the languages. That is, the structure needs 16 dimensions, so its ratio to the hidden width, which we call **capacity**, runs from 0.125 to 2.0. Below one there is room to give every component its own axis, so any entanglement was chosen; above one there are more components

than dimensions to put them in, so they cannot all have their own axis and entanglement is forced (Elhage et al., 2022).

2.2 Method

2.2.1 The learner

The input to the learner model comprises three pieces of information: the form, the class, and the context. Each has its own embedding, and the three are concatenated and passed to a standard multi-layer perceptron (MLP) with a single hidden layer of width d and either a linear or a ReLU activation function, which outputs a distribution over the vocabulary (Figure 1). The training objective is thus to minimize cross-entropy between the model’s predicted distribution over the vocabulary and the true distribution.

2.2.2 Decomposing a representation

We measure separability by decomposing a word-in-context representation into its lexical representation, its class representation, and its idiosyncratic representation. The representation is the MLP’s hidden layer, which for a given form and class varies across the contexts it is observed in, so let $M[i, c] \in \mathbb{R}^d$ be the mean of those hidden states for item i used at class c . We decompose that grid by centering, so that

$$M[i, c] = \mu + \alpha(i) + \beta(c) + \gamma(i, c), \quad (2)$$

where μ is the grand mean; $\alpha(i)$, item i ’s mean across classes minus μ , carries the lexical information; $\beta(c)$, the mean across items at class c minus μ , carries the class information; and $\gamma(i, c)$ carries the idiosyncratic information, what item i at class c holds that neither marginal accounts for (Figure 2). Because every item is observed in every class with equal weight, neither marginal absorbs part of the other. The four terms add back to $M[i, c]$ exactly, and no two of them vary together across the grid, so their squared lengths sum to the total and we report each as a share of it.

Each of these averages is taken over finitely many observations, which differ only in context, so each carries some of the context term of Equation 1 as noise. For γ this matters: a squared length is always positive, so its size would come out above zero even for a language with no idiosyncratic information in it at all. We therefore divide the observations behind each average into two even-sized

splits, run the decomposition on each split separately, and take the dot product of the two estimates of γ , $\langle \gamma_A, \gamma_B \rangle$, which is large when the splits agree about the idiosyncratic information and near zero when they do not. It is the contexts that are divided and not the words, so every word appears in both splits, and the dot product compares two estimates of the same word built from different sentences. Noise that differs between the splits cancels, and unlike a squared length this can come out negative, so zero is a value it can take rather than a floor it sits above.

For each representation we report two measurements. The first measures the fraction of the representation accounted for by lexical, class and idiosyncratic information: each component’s squared length, summed over its vectors and divided by the total, so that the three fractions sum to one. The second measures whether those components occupy the same directions. In order to do so, we project the item vectors onto the class direction and record the fraction of their squared length that survives, and we project the interaction onto the smallest subspace containing both the item and the class vectors (i.e., the span of the two subspaces). A representation is separable if a word’s representation is the sum of an item part and a class part with the two on their own axes, so it fails either when the components are non-orthogonal, so that projecting out the class also removes item information, or when there is a third component that is neither an item part nor a class part, whatever its orientation. The question is accordingly not whether class information is present, but how it is carried: on a single direction shared across words, or on a separate direction for each of them.

In order to quantify the effect of size, we compare each to a random permutation distribution. For class, we permute the class labels within each item: for a given word, whether a representation is labeled as a noun or as a verb is decided at random, independently for every word. Each word keeps both of its representations, so the item effect is untouched, but any systematic difference between the two classes is averaged away. For item and for the interaction, both measured as the agreement between two splits of the same words, we permute which item of one split is paired with which item of the other. Each split is left exactly as measured, but each word is now matched with a different word instead of with its own second estimate, so there is no shared structure left for the comparison to find.

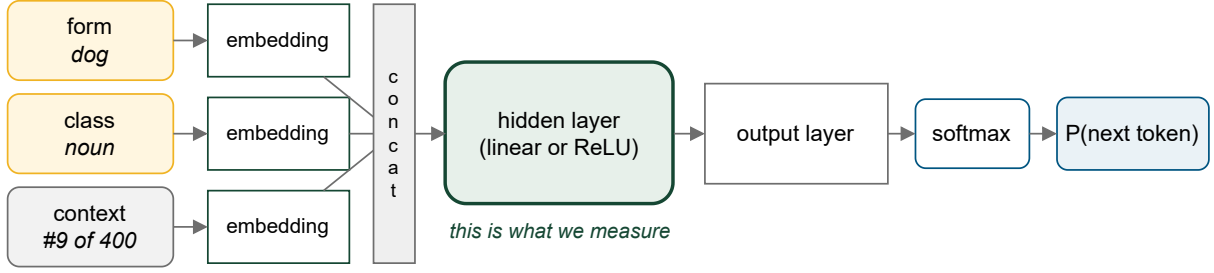


Figure 1: What the learner sees. Every form is observed 24 times at each class, each time in a different context; that variation across contexts is what the measurement later splits in two.

For example, instead of matching *walk*’s representations from one split with *walk*’s from the other, we match them with some other word’s, such as *run*, so that what we measure is the agreement two unrelated words happen to show by chance. We then recompute the same quantities we computed from Equation 2, two hundred times for each. The resulting distribution is what the size would be if that structure were absent, and we report how many measurements fall outside its 95% interval.

In order to quantify whether the overlap is meaningful, we measure the overlap an effect has with a random subspace. We do so by holding the effect exactly as measured and projecting it onto a random subspace of the same rank, two hundred times, which leaves the effect untouched and varies only the subspace. These two hundred projections give us a distribution of overlaps that an unrelated subspace would yield. An effect lower than this baseline indicates meaningful orthogonality; an effect larger than it indicates meaningful non-orthogonality. We present each overlap as a multiple of the median of this distribution, together with how many measurements fall outside its 95% interval. This gives both the size of the effect and how consistently it holds across measurements.

2.3 Results

The weights in Equation 1 describe the true distribution the model is trained to match, while the measure reads the representation it builds, so their numbers do not match: a learner handed a language that is 85% interaction can build a representation in which the interaction is a far smaller share, because the representation only has to be enough to produce the answer. Validation asks that the measure track what was planted, read zero when zero was planted, and have a known floor. Appendix A reports the separate check in which the components are planted directly in a representation.

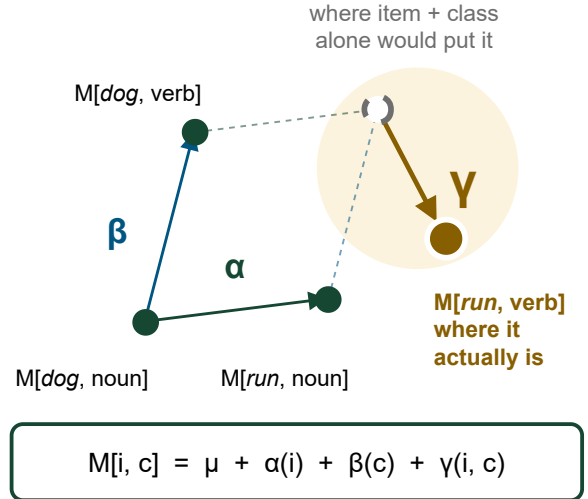


Figure 2: If a word’s representation were an item part plus a class part, three corners of a 2×2 grid of means would fix the fourth and the parallelogram would close. It does not. γ is how far the actual mean sits from where that additive account puts it, and it is not error: the four components exhaust the grid exactly, so γ is the part that belongs to the pairing of a particular word with a particular class. The arrow’s length is $\|\gamma(i, c)\|$, and with two items and two classes it is the whole interaction, since γ then has a single degree of freedom and the four means share one vector up to sign. What we report is each component’s squared length as a share of the structure across the grid.

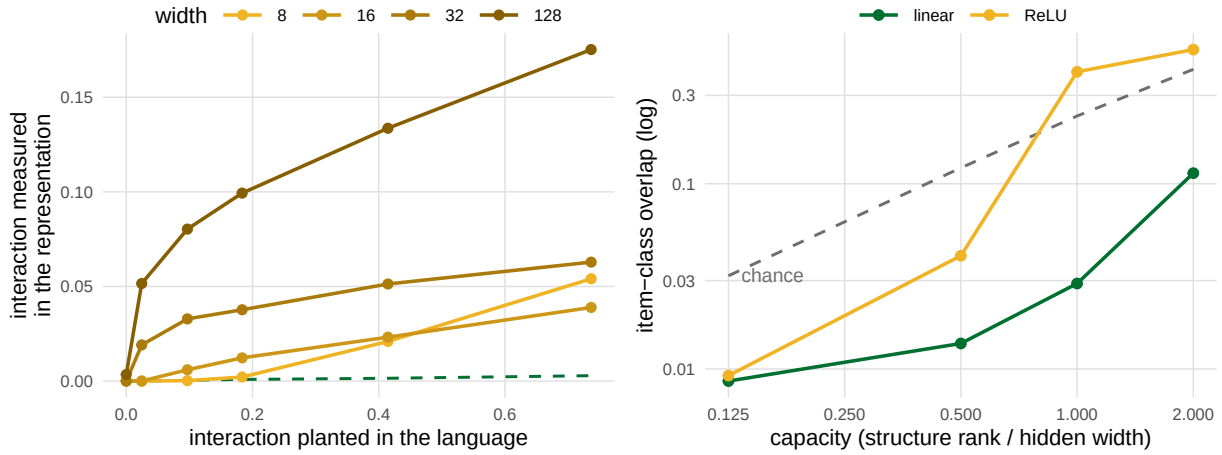


Figure 3: What the artificial-language grid settles about the measure and about the learner, over all 7,560 runs. Left: the measured interaction against the amount planted in the language, one line per hidden width, with the linear condition dashed. The linear condition cannot represent an interaction whatever the language contains and reads zero at every width; under ReLU the measured share rises with the planted one, and the wider the representation the more of it is expressed. The two axes are not on a common scale: the planted share is a share of the target distribution and the measured share is a share of the representation the model builds, and Appendix A shows the measure recovers a planted share exactly when the planting is done in a representation. Right: overlap between the item and class effects against capacity, with the dashed line marking the upper end of the null for an arbitrarily oriented effect. Below capacity the two effects are more orthogonal than chance; once dimension counting makes separation impossible they are several times chance. Note the log axis, and that the floor moves by a factor of eight across these widths.

The measure reports zero when the answer is zero. Under the linear activation the hidden layer is a weighted sum of the three embeddings, so its interaction is identically zero however much the language contains (Appendix B), and every reading the measure gives there is its own noise. Across all 2,880 runs where the language *did* contain an interaction, it reports a median share of 0.0010 and flags one in 5.0% of them, the nominal rate for a test at the 5% level; where no interaction was planted it flags 4.9%. This calibration comes from a condition whose true value is provably zero, measured at every width, sample size and noise level in the grid.

The measure also tracks what is there. Under ReLU the measured interaction rises monotonically with the planted one (Figure 3, left), and the rate at which it is flagged rises with it, from 38% at the smallest nonzero setting to 96% at the largest. With no interaction planted, ReLU flags 2.9% of runs, below nominal: the nonlinearity does not manufacture an interaction out of an additive language. The measured shares are smaller than the planted ones throughout, and the gap is a fact about the representation rather than about the measure: planting the same components directly in a representation recovers them almost exactly in every class (Ap-

pendix A), so a model trained on a language built largely out of the interaction still builds a representation that carries much less of it. How much less depends on how wide the representation is, and what matters here is that the relationship is monotone and that zero maps to zero.

Item and class effects entangle only when there is no room for them not to, and this is the substantive result (Figure 3, right; Table 1). With capacity well below one, ReLU’s item-class overlap is 0.0092 against a chance level of 0.0295, so the two effects are *more* orthogonal than two arbitrary directions would be, which is what a learner that has room and uses it looks like. At capacity 1 the overlap is 0.402 against a floor of 0.2577, about 1.5 times chance, and at capacity 2 it is 0.530; the linear condition stays at or below chance throughout. This is superposition in a setting where we can watch it happen: when the number of directions the structure needs exceeds the number available, the components stop being orthogonal because they cannot all be, and nothing about the language changed across those runs.

The measure also works when the representation primarily encodes contextual information unrelated to the specific form and class. When the context term accounts for more than 80% of the variance

condition	capacity	runs	interaction	item-class overlap	interaction overlap
linear	0.125	405	0.005 (235/405)	0.0086 (0.4x, 332/405)	
linear	0.500	405	0.002 (183/405)	0.0137 (0.2x, 400/405)	
linear	1.000	405	0.000 (98/405)	0.0289 (0.2x, 349/393)	
linear	2.000	405	0.000 (116/405)	0.1140 (0.5x, 196/332)	
ReLU	0.125	405	0.080 (405/405)	0.0092 (0.6x, 221/405)	0.087 (1.3x, 493/405)
ReLU	0.500	405	0.035 (366/405)	0.0408 (0.7x, 236/405)	0.142 (0.7x, 202/324)
ReLU	1.000	405	0.007 (229/405)	0.4022 (7.1x, 275/394)	0.227 (1.0x, 357/344)
ReLU	2.000	405	0.001 (168/405)	0.5303 (5.6x, 125/231)	0.538 (2.0x, 166/160)

Table 1: What a learner builds from the same languages as its representation runs out of room. Capacity is the rank the structure needs divided by the hidden width, so the rows run from a width of 128 down to 8 while the languages are held fixed: the median interaction planted is 0.097 in every row, and the fall down the *interaction* column is therefore the width and not the language. *Interaction* is the interaction’s share of the representation, a size. The last two columns are orientations, and a separate question: how much of the item effect lies in the class subspace, and how much of the interaction lies in the span of the two main effects. Each is the fraction that survives the projection, which is the quantity that matters, given as the median, then in parentheses how many times its own null’s median it is and how many of the runs in that row fall outside that null’s 95% interval, over the number of runs in the *runs* column. The null holds the effect fixed and rotates the target subspace at random 200 times. Reporting the multiple rather than the raw fraction is what makes the rows comparable: the null itself moves several-fold down this table as the width falls, so the fractions are not comparable across rows on their own. A value below one is more orthogonal than an arbitrary orientation would be, which with room to spare ReLU is. Table ef{tbl-llm} reports the same three things for the language models. The linear arm has no interaction to orient, so its last column is empty. Below capacity one a fully separable solution is reachable by dimension counting; above one it is not.

in the logits, leaving under a fifth for the form, the class and their interaction combined, ReLU’s measured interaction holds at 0.021 against 0.038 when context is minor, and the flag rate falls from 70% to 44%. Power degrades, as it must with twenty-four observations per item and class, but the estimate does not drift. That is the regime Experiment 2 sits in, since a word’s hidden state varies mostly with the sentence it appears in rather than with the word or its class.

Two things carry into Experiment 2. The instrument reads zero when the answer is zero and tracks the answer when it is not, at the sample size and noise level the language models will be measured at. And whether the item and class effects come out orthogonal is governed by whether the representation had room, which is a fact about capacity rather than about the language, so near-orthogonality in a wide model is uninformative on its own and has to be read against its own chance level.

3 Experiment 2: Item and Category in Language Models

Experiment 1 established that a learner fitting an interactive language cannot represent it as item plus class, and that a nonlinearity additionally makes the components non-orthogonal. Experiment 2 applies the same decomposition to the contextual representations of pretrained language models, across three linguistic distinctions chosen to span grammar and meaning.

3.1 Dataset

For part of speech and grammatical role we concatenate six general-domain English Universal Dependencies treebanks (Nivre et al., 2020): EWT, GUM, LinES, ParTUT, GENTLE and PUD, giving 41,095 sentences and 706,323 tokens. A lemma must be attested in both classes often enough to estimate a mean at each, and a single treebank yields too few such lemmas. For metaphor we use the VUA20 corpus (Leong et al., 2020; Steen et al., 2010), restricted to content words, which supplies 88,509 labeled tokens.

All three constructions are **same-token**: the item’s surface form is identical across the two classes, so the representation is read from the same token in both, and no part of any difference can come from tokenization.

Keying the item on the *lemma* instead admits more items (130 rather than 78, drawn from 999

distinction	levels	analyzed	tokens
part of speech	NOUN / VERB	78	4,722
grammatical role	<i>nsubj</i> / <i>obj</i>	49	2,150
metaphor	literal / figurative	180	14,871

Table 2: The three constructions. All three key the item on the **surface form**, so the token is identical in both classes of the distinction. *Analyzed* counts the forms attested often enough at each level to estimate a mean at each.

lemmas attested as both), but a lemma pools every inflected form: *run* and *runs* as a noun against *run*, *runs*, *ran* and *running* as a verb. Those are different tokens, and the consequence shows up at the embedding layer, which has no context to work with and so can only distinguish classes that differ in their input. Keyed on the lemma the layer-0 interaction is 0.083 to 0.172 across models; keyed on the form it is 0.0010 to 0.0028. The difference is inflection rather than representation, and keying on the form removes it at the cost of a smaller item set. Appendix C shows what the measure does on a distinction where the property cannot be arranged at all.

Identifiability. The interaction $\gamma(i, c)$ is estimable only for an item observed in more than one class, since it is defined by how an item’s representation shifts across classes, and we additionally require at least 10 tokens for each item in each class so that a mean can be estimated and then split in two. This restricts the scope for part of speech, where it selects forms genuinely used both ways, such as *run* or *hit*. Role and metaphor carry no such restriction, since any noun occurs as both subject and object and any content word occurs both literally and figuratively, which is one reason they broaden the claim. The median item holds around 24 tokens in each class across the three constructions, the sample size the measure is validated at in Appendix A.

We measure two model families at three matched scales: OPT-BabyLM, trained on developmentally plausible quantities of child-available language (Warstadt et al., 2023), and Pythia, trained on the Pile (Biderman et al., 2023), at 125M/160M, 350M/410M and 1.3B/1.4B parameters, with widths of 768, 1024 and 2048. Appendix D gives parameter counts and training-token estimates.

3.2 Method

The decomposition, the two reported quantities and the replication logic are those of Section 2.2. For every token we take the hidden state of the word’s **last subword** at every layer, from the embeddings through the final block, aligning subwords to corpus tokens with the tokenizer’s word alignment. Sentences are drawn in a fixed random order and processed until roughly 300K tokens are extracted per model, so all models see the same material, and representations are extracted once with every analysis running offline from them.

The two independent estimates come from the corpus itself: each item in a class is a word used across many sentences, so we divide its token contexts into two disjoint, even-sized splits and estimate the grid from each. What cancels is context noise, the variation in a word’s hidden state across the sentences it appears in, which is the same thing the split cancels in Experiment 1.

Standardization. Transformer hidden states contain a few *rogue* dimensions whose variance exceeds the rest by two to three orders of magnitude (Timkey and van Schijndel, 2021), and these would otherwise dominate any subspace estimated from the raw scatter, so we divide each dimension by its standard deviation before decomposing. This is a diagonal rescaling rather than a whitening, so it needs no favorable ratio of samples to dimensions; the measure is accordingly invariant to per-dimension rescaling but not to arbitrary linear reparameterization, which we return to in Appendix B.

3.3 Results

The two main effects sit on separate axes. At the deepest layer, item-class overlap runs 0.0032 to 0.0067 for part of speech and 0.0025 to 0.0062 for grammatical role, so projecting out the class direction leaves better than 99% of the item effect intact. Both sit above the upper end of their null, so the residual alignment is real rather than a by-product of the dimension count, but it is small enough that these representations do not achieve class structure by borrowing the directions that carry word identity. Where a shared class direction exists, it is its own direction.

The interaction, however, does not have axes of its own, and this is what decides whether a third component could simply be added to the description and separability recovered in a wider sense.

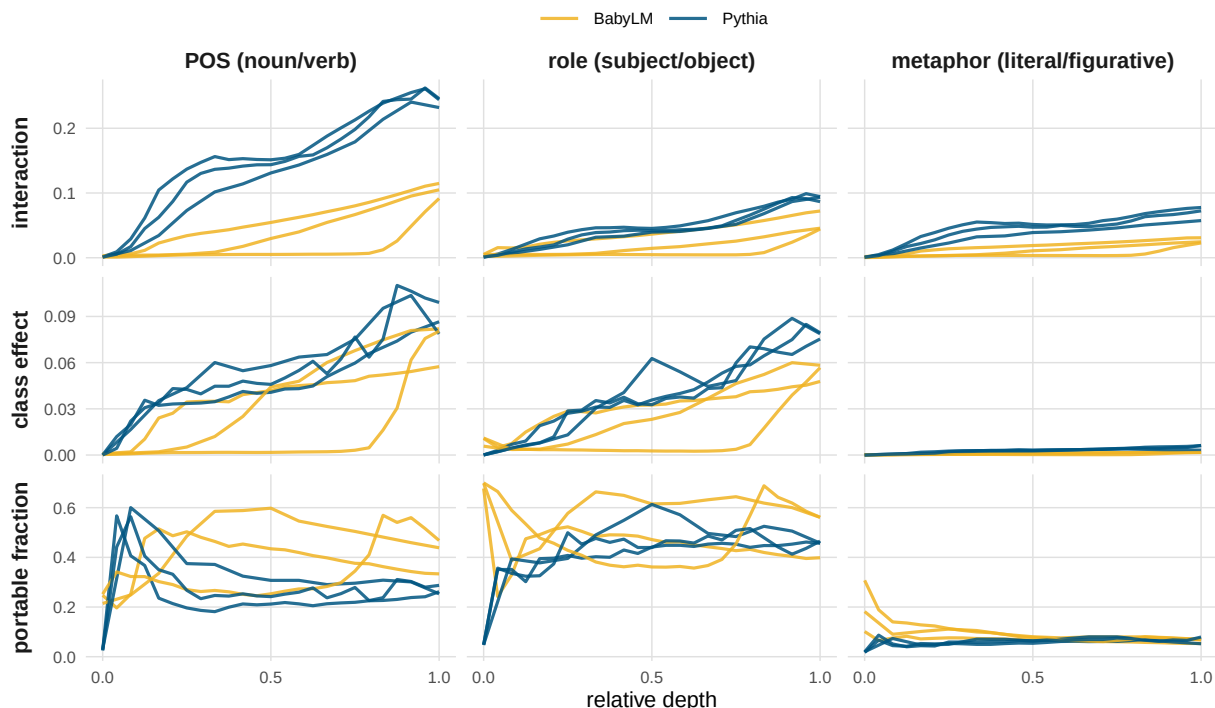


Figure 4: How the three components grow through the network, for every model and every distinction. Relative depth runs from the static embedding layer (0) to the final block (1). Top: the interaction’s share of representational structure. Middle: the class effect’s share. Bottom: the portable fraction, the class effect’s share of the two together, which is the quantity the three distinctions are ordered by. Each line is one model; color distinguishes the two training corpora. Both components are built by depth, and the interaction is the larger of the two throughout. The deepest-layer values, with their chance levels, are in Table 11.

For part of speech 0.191 of the interaction lies in the span of the two main effects, for role 0.131 and for metaphor 0.227, against nulls whose upper ends are 0.040, 0.026 and 0.040 (Table 11). The partition of the grid says nothing about direction, so these numbers are not implied by the decomposition: the item-by-class term reuses the very directions that carry item identity and class membership. Read with the first overlap, they say something more specific than “not separable”. The failure is not that item and class have been collapsed onto shared axes, since those remain distinct; it is that the part of class structure specific to each word is written in the same directions as everything else. For metaphor the question does not arise, because the class effect never rises above the level at which an orientation is worth reporting, which is itself the first indication of how little of metaphoricity is carried by a shared direction. It is not, however, absent. Against its own permutation distribution the class effect falls outside the 95% interval in 6 of 6 models, so metaphoricity does have a direction shared across words; it is simply a very small one. What separates the three constructions is not whether they clear their permutation distri-

butions, which all of them do, but by how much: the class effect is 29.4 times its null’s median for part of speech and 23.7 times for grammatical role, against 5.5 times for metaphor (Table 11). A class effect this small is exactly the case a size cannot settle on its own, since a squared length sits above a floor set by how many observations went into it, and the permutation is what distinguishes a small effect from no effect.

A substantial interaction is there, and depth builds it. In all three distinctions the interaction is essentially absent at layer 0, between 0.0002 and 0.0051 across every construction and model, and rises monotonically with depth to between 0.023 and 0.246 (Figure 4, top row). Because the item is keyed on the surface form, the token is identical in both classes, so the static embedding layer has nothing to distinguish them with, and it does not. Everything the measure finds is therefore created by contextualization rather than inherited from the input, and it is not positional: zeroing OPT-BabyLM’s absolute position embeddings leaves the class structure more word-specific rather than less (Appendix F). This is the component that breaks separability, since it is neither a fact about the word

nor a fact about the class.

All three sizes are shares of the grid, and the grid has already averaged away how a word varies across the contexts it appears in, so how much of the representation the grid is in the first place is worth stating. At the embedding layer it is 0.83 to 0.85 of the variance across the three constructions, which is the same-token design working as intended: identical tokens differ little before anything contextualizes them. At the deepest layer it is 0.32 to 0.33, so most of what the model computes there is about the sentence a word is in rather than about the word and its class. Multiplying the two through, the deepest-layer item effect for part of speech is 0.25 of the representation rather than the 0.75 it is of the grid, and the class effect and interaction scale down in the same proportion, with context taking the remainder. Both figures are corrected for the noise each grid entry carries as an average over finitely many observations, which otherwise inflates the grid’s apparent share by an amount that shrinks as tokens per item grow and so differs across constructions (Appendix B).

The class effect grows with depth as well (Figure 4, middle row), and that matters for the framing. Models are not choosing between abstraction and entanglement; contextualization builds both at once. A description of these representations as either abstract or exemplar-like is therefore underdetermined by the data, and the useful question is not which one they are but what the balance is.

That balance is graded across distinctions. Table 11 reports it as the *portable fraction*: of the structure that distinguishes the two classes, how much is a direction shared across words rather than specific to each word. Grammatical role is the most portable at 0.46, part of speech next at 0.31, and metaphor by far the least at 0.07, so only about a fourteenth of metaphor’s class-relevant structure is carried by a shared direction. The ordering is worth noting: part of speech has the largest class effect of the three in absolute terms, but also much the largest interaction, so proportionally less of what distinguishes its classes is portable than for grammatical role. Metaphoricity is close to being encoded word by word, and what makes a use of *plant* figurative is largely not what makes a use of *course* figurative.

4 Experiment 3: Does the Interaction Carry the Class?

Experiment 2 shows that the interaction is large and that it replicates across independent splits of a word’s occurrences, but neither shows what it is *about*. An item-by-class term captures any reliable difference between a word’s uses in the two classes, and some of that difference is context correlated with the label rather than the label itself, since metaphorical uses occur in different sentences from literal ones and subjects appear in different constructions from objects. So γ is an upper bound on class-binding, and the question is whether any of it is the class.

4.1 Method

We test it by decoding. If γ encodes the class, the class should be recoverable from the part of a token’s representation that is neither the item effect nor the shared class direction, and we remove both explicitly. All decoding happens *within* a single word, so we ask whether *plant*’s figurative uses can be told from *plant*’s literal ones, never whether *plant* can be told from *account*; the item effect is constant within a word and so cannot contribute. We then estimate β as the difference between the two class means pooled over all items, normalize it, and remove that component from every token, so that whatever separates the classes afterwards is not the direction they share.

What remains is the interaction: the part of a word-at-a-class that is neither a fact about the word nor a fact about the class. We decode it with a cross-validated logistic probe, five folds, classes balanced by subsampling within each word so that chance is 0.50, and average the resulting accuracies over words, with a word contributing only if it has at least eight tokens in each class. Accuracies are averaged unweighted over words and some words rest on few tokens, so the mean carries a percentile bootstrap interval over words. The whole procedure is then repeated with the class labels permuted within each word, which preserves every word’s token count and class balance while destroying the labelling, and that null is where a probe reading its own cross-validation folds would land. We run this at the deepest layer of the largest model in each family, for all three constructions.

distinction	family	words	class	null
part of speech	BabyLM	78	0.93 [0.91, 0.94]	0.52 [0.49, 0.55]
part of speech	Pythia	78	0.94 [0.93, 0.95]	0.50 [0.48, 0.53]
grammatical role	BabyLM	49	0.85 [0.82, 0.87]	0.50 [0.47, 0.53]
grammatical role	Pythia	49	0.84 [0.82, 0.87]	0.52 [0.49, 0.55]
metaphor	BabyLM	180	0.65 [0.63, 0.66]	0.49 [0.47, 0.50]
metaphor	Pythia	180	0.70 [0.68, 0.71]	0.49 [0.48, 0.51]

Table 3: Class decoded from the interaction, at the deepest layer of the largest model in each family. The item is held fixed and the shared class direction is projected out, so neither the item effect nor the class effect can contribute. *Class* is the accuracy of that decode and *null* permutes class labels within each word; brackets are percentile bootstrap intervals over words. Chance is 0.50.

4.2 Results

The interaction is about the class. In every construction and both families the class is recoverable from the interaction at well above chance while the permuted null sits on chance (Table 3), reaching 0.94 for part of speech, 0.84 for grammatical role and 0.70 for metaphor, with bootstrap intervals excluding the nulls by a wide margin in all six cases. The null landing on chance is what makes those accuracies interpretable, since it is the same probe, the same folds, the same word-level averaging and the same small per-word samples with only the labelling destroyed.

The effect is graded the same way the interaction’s size is. Part of speech decodes best, role next, metaphor least, which is the ordering of the portable fraction in Table 11 read backwards: the distinctions whose class-relevant structure is least portable are also the ones whose interaction carries the class least strongly. Two measurements, one geometric and one functional, agree on the same ranking.

Metaphor is the case where this matters most. Its class effect never clears the threshold at which an orientation is worth reporting (Section 3.3), so on the geometric evidence alone one could conclude that metaphoricity is not represented at all, yet it decodes from the interaction at 0.70 against a null of 0.49. It is represented; it is simply not represented as a direction shared across words, and a method that looks only for a shared direction would have missed it.

This result has a definite scope. It shows that γ carries the specific label rather than generic correlated context, since generic context would not survive holding the word fixed and removing the

shared axis. It does not show that γ is *only* the label: context correlated with the class within a word survives both removals, so the interaction remains an upper bound on class-binding, and this raises the floor rather than closing the gap.

5 Discussion

Limitations

Everything here is English, and rests on three distinctions measured in two model families at three scales each. The graded ordering we report is an ordering across *these* distinctions, and whether it holds for aspect, animacy or definiteness is untested.

The measure is invariant to per-dimension rescaling and to rotation, but not to arbitrary linear reparameterization, so overlap between the item and class effects is relative to the model’s own basis. We take that to be the basis of interest, since LayerNorm and the feed-forward nonlinearities act coordinate-wise and a downstream output layer operates on the unwhitened space; a reader who wants a fully gauge-free quantity will not find one here. The component sizes, their growth with depth and their significance do not depend on that choice.

Two things we would like to have done. These are endpoint measurements, so they say nothing about the order in which the two components are learned, which needs training checkpoints. And the interaction captures any systematic within-word difference between the classes, so a distinction realized through the contexts a word appears in is measured together with those contexts. For a contextual model we take that to be what representing the distinction consists in, but separating the two would need a construction in which they can be varied independently.

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A Validating the Measure

Experiment 1 plants structure in a *language* and measures a trained model’s representation. A discrepancy there is ambiguous between the measure missing something and the model not mirroring the language, and the second is known to happen: the linear condition cannot represent an interaction at all. This appendix removes the ambiguity by planting μ , α , β and γ **directly in a representation**, with no model and no training, and passing the result to the same function the language models are measured with. Ground truth now sits at the level the measure operates on.

Construction. The three components are drawn at random and then centered so each satisfies the constraints its role implies: α and β sum to zero over their own index, γ over both. Without that a planted γ would partly be an item or class main effect and the planted shares would be wrong before the measure saw them. Each is rescaled to hit a target share of the total squared length across the grid, under the same weighting the measure uses. Each mean is then observed n times under isotropic noise, whose scale is set as a multiple of the mean squared length. The grid crosses the planted shares (the same factorial Experiment 1 runs, mapped into share space, plus the three compositions Experiment 2 measures), item counts from 27 to 180, two and four classes, widths from 8 to 2048, 10, 24 and 100 observations per item and class, three noise levels, planted-orthogonal and planted-non-orthogonal effects, and five seeds: 252,000 runs.

Recovery. At Experiment 2’s regime, 24 observations per item and class in 768 or more dimensions, the measured interaction matches the planted one across the whole range (Table 4), and is flagged in essentially every run where anything was planted and none where nothing was.

planted	runs	measured	flagged
0.000	6300	0.0000 [0.0000, 0.0074]	0%
0.030	900	0.0323 [0.0244, 0.0372]	98%
0.048	1800	0.0494 [0.0413, 0.0680]	100%
0.056	1800	0.0571 [0.0493, 0.0907]	100%
0.059	1800	0.0605 [0.0526, 0.1024]	100%
0.100	1800	0.1006 [0.0921, 0.1051]	100%
0.111	900	0.1149 [0.1035, 0.1274]	100%
0.167	1800	0.1701 [0.1582, 0.2188]	100%
0.190	1800	0.1926 [0.1817, 0.2709]	100%
0.200	1800	0.2020 [0.1910, 0.2950]	100%
0.240	900	0.2407 [0.2284, 0.2445]	100%
0.333	900	0.3372 [0.3203, 0.3603]	100%

Table 4: Recovery at 24 observations per item and class in 768 or more dimensions, pooled over noise levels, item counts and widths. Brackets are the 5th and 95th percentiles across runs.

Floors. The interaction and the item effect read essentially zero when nothing is planted, at every sample size. The class effect does not, and this is the one quantity in the paper that carries a floor worth knowing (Table 5): at 24 observations per item and class a true zero reads 0.0036. This is the price of estimating the class effect as a squared length rather than a cross-product, and it is why the metaphor result in Section 3.3 is stated as an inability to distinguish the class effect from zero rather than as a measurement of something small. Metaphor’s observed class effect, 0.0016 to 0.0062 at the deepest layer, sits at that floor.

effect	observations	median	95th pct	flagged
class effect	10	0.00836	0.0849	—
class effect	24	0.00359	0.0365	—
class effect	100	0.00085	0.0093	—
interaction	10	0.00000	0.0557	2.6%
interaction	24	0.00002	0.0312	1.4%
interaction	100	0.00003	0.0084	0.4%
item effect	10	0.00008	0.0559	—
item effect	24	0.00002	0.0232	—

(continued)

effect	observations	median	95th pct	flagged
item effect	100	0.00000	0.0068	—

Table 5: What the measure reports when an effect is planted at exactly zero, by observations per item and class, pooled over every other axis. The interaction is flagged well below the nominal 5%; the class effect is not cross-denoised and carries a floor that shrinks with sample size.

The chance level for overlap. With the item and class effects planted orthogonal, the measured overlap tracks r/d closely and sits a little below it (Table 6). This is the number Experiment 2’s overlaps are read against: at $d = 2048$ a genuinely orthogonal pair reads 0.00102, and the values reported in Section 3.3 are two to seven times that.

width d	measured	analytic chance
8	0.19137	0.25000
16	0.10758	0.12500
32	0.05613	0.06250
128	0.01813	0.02344
768	0.00290	0.00391
1024	0.00228	0.00293
2048	0.00102	0.00146

Table 6: Measured item-class overlap when the two effects are planted orthogonal, at 24 observations per item and class. The *chance* column is the analytic level of Appendix B.

B The Decomposition

B.1 The grid

A word used in a given class is observed in many contexts, and the representation differs a little in each. Averaging those observations within the pairing gives one entry of the grid: for items $i \in \{1, \dots, I\}$ crossed with classes $c \in \{1, \dots, C\}$, $M[i, c] \in \mathbb{R}^d$ is the mean representation of item i in class c (Figure ??). An item observed in only one class is dropped, as is an item with fewer than 10 tokens in either class, so the grid is complete and every entry rests on a comparable number of observations. The variation across contexts that the averaging sets aside is not discarded; it is what the two splits of Section 2.2.2 are built from.

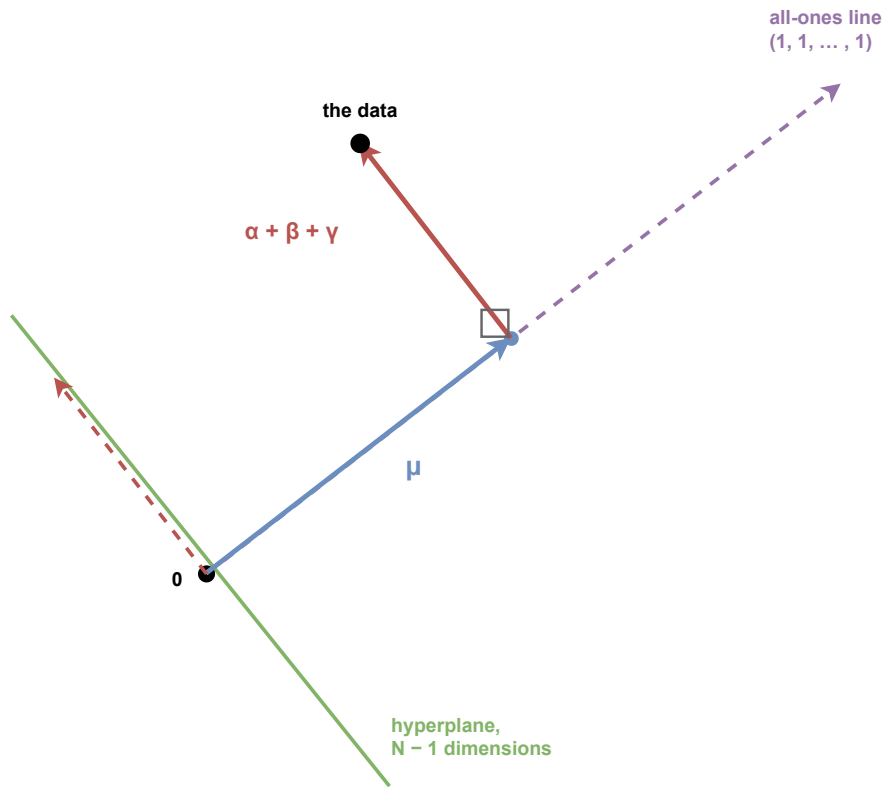


Figure 6: Centering, drawn for a single hidden dimension read across the whole grid. The all-ones direction is where every position takes the same value, so it carries no differences and no variance, and μ is the projection onto it. Removing it leaves a hyperplane of dimension $N - 1$, and every remaining component is a displacement inside it. The dashed copy is that displacement moved to the origin, where it lies in the hyperplane.

Two spaces are in play. The representations themselves live in $\mathbb{R}^d$. The decomposition happens in the grid index space: fix one hidden dimension, read it across all IC positions of the grid, and it is a vector in $\mathbb{R}^{IC}$ (Figure 6). Everything below is that operation carried out once per hidden dimension, with the results stacked back up into the vectors α_i , β_c and γ_{ic} .

In that space the all-ones direction is the one in which every position of the grid takes the same value. Moving along it adds the same amount everywhere and creates no difference between any two positions, so it is the direction along which nothing distinguishes anything, and μ is the projection onto it. Removing it leaves the orthogonal complement, and the three remaining components are displacements inside that complement rather than raw values.

Each component is fixed by two choices, one per axis: along the item axis either average or center, and along the class axis the same. Averaging keeps what is common and discards what makes positions differ along that axis; centering does the reverse. The four combinations are the four components of Equation 2:

	varies over items	varies over classes
μ	no	no
α_i	yes	no
β_c	no	yes
γ_{ic}	yes	yes

So α_i is what holds of an item in both classes, β_c is what holds of a class for every item, and γ_{ic} is what varies along both at once, which is why it cannot be a fact about either one alone. Since averaging

and centering along an axis add back to what was there, the four reconstruct $M[i, c]$ exactly, and they are mutually orthogonal.

Centering an axis costs one dimension, the all-ones direction along it. The four components therefore occupy 1, $I-1$, $C-1$ and $(I-1)(C-1)$ of the grid's IC positions, which is why two classes leave β a single direction whatever I is. Nothing is left over as error: γ is not a residual in the sense of a model that failed to fit, but the part of the representation belonging to the pairing of a particular item with a particular class.

Two consequences are used in the body. First, $\sum_i \alpha_i = 0$, $\sum_c \beta_c = 0$ and $\sum_i \gamma_{ic} = \sum_c \gamma_{ic} = 0$ follow from the centering rather than being imposed as constraints. Second, an **additive** grid, one of the form $M[i, c] = A(i) + B(c)$, has $\gamma = 0$ *identically*: writing $\bar{A}$ and $\bar{B}$ for the averages, $\mu = \bar{A} + \bar{B}$, $\alpha_i = A(i) - \bar{A}$, $\beta_c = B(c) - \bar{B}$, and the remainder cancels exactly. This is what makes the linear condition of Experiment 1 a negative control rather than an assumption.

B.3 Sizes

Because the three components are mutually orthogonal, their squared lengths add rather than overlapping, and the total squared length across the grid partitions exactly:

$$\sum_{i,c} \|M[i, c] - \mu\|^2 = C \sum_i \|\alpha_i\|^2 + I \sum_c \|\beta_c\|^2 + \sum_{i,c} \|\gamma_{ic}\|^2,$$

C and I are not a choice of scaling. They are how many positions of the grid each component occupies (Figure ??): α_i does not depend on the class, so it sits at all C positions of its row, and β_c does not depend on the item, so it sits at all I positions of its column. Every one of the I items is displaced by the same β_c , so the displacement present in the grid is I copies of it. The interaction is simply summed because it takes its own value at every position and is never repeated. Dividing all three by IC would report averages per position instead of totals and leave every share unchanged; what the occupancy counts rule out is weighting the three differently from one another.

Sizes are reported as shares of the total. All three terms scale linearly in I , so the shares do not depend on how many items the grid contains and are comparable across constructions of different size.

What the grid is a share of. The partition above is of the grid, which has already averaged the contexts away, so it leaves open how much of the representation the grid is. Splitting each observation into its pairing mean and its deviation from it, and using that deviations from a mean sum to zero, gives a second exact partition one level up:

$$\sum_x \|x - \bar{x}\|^2 = \underbrace{\sum_{i,c} \sum_{x \in (i,c)} \|x - \hat{M}[i, c]\|^2}_{\text{within a pairing}} + \underbrace{\sum_{i,c} n_{ic} \|\hat{M}[i, c] - \bar{x}\|^2}_{\text{between pairings}}.$$

The raw between fraction overstates the grid, because $\hat{M}[i, c]$ is itself an average of n_{ic} observations and carries $d\sigma^2/n_{ic}$ of context noise that lands in the between term. The bias falls as tokens per item rise, so it differs across constructions and would not be comparable between them. We therefore report the variance-component form: the within mean square estimates the context variance, the between mean square estimates that plus n_0 times the true spread of the pairing means with n_0 the effective group size of the unbalanced design, and the difference gives the spread alone. Planting no structure at all, the raw fraction reads 0.081 at 12 observations per pairing and 0.005 at 200, while the corrected one reads zero at both.

Because the corrected quantities are variance components rather than weighted sums, they compose with the sizes: each size times the grid's share gives that component's share of the representation, and the three plus the context term account for it entirely.

B.4 The overlaps and their chance level

The sizes say how much of the representation each component accounts for and nothing about direction. Two components can both be large whether or not they occupy the same directions, and separability is the

second question: a word’s representation is an item part plus a class part only if the two can be varied independently, which requires them to lie on distinct axes. That question lives in $\mathbb{R}^d$, not in the grid index space the sizes were computed in, and unlike the sizes it does not partition, because each overlap is a relation between two particular components rather than a share of one whole.

Each overlap projects one component into the subspace another occupies:

projected into the subspace of	
α	β
β	α
γ	α and β jointly

The first two are not redundant with each other. One direction and thirty-odd are very different targets to land in by chance, and the chance level below is what puts them on a comparable footing.

The subspace. Subspaces are orthonormal bases from the leading right singular vectors, with rank set by the participation ratio $r = \text{round}((\sum_k \lambda_k)^2 / \sum_k \lambda_k^2)$ of the scatter eigenvalues, capped at the algebraic maxima above. The participation ratio counts directions discounted by how much each carries: it returns 1 when all the spread lies along one direction and k when it is even across k of them. With two classes β spans one direction, so $r_{\text{class}} = 1$ throughout Experiment 2.

The projection. Writing S for the $d \times r$ basis and V for the component being projected, the overlap is $\|VS\|_F^2 / \|V\|_F^2$. V is complete, every vector of the component at full dimension; only S is rank-truncated. So the rank enters through the target alone, which is why it fixes the chance level without altering any measured overlap.

It is a squared cosine. For one vector and one direction, $\langle v, u \rangle^2 / \|v\|^2 = \cos^2 \theta$ exactly, so the ratio is a length-weighted mean of $\cos^2 \theta_i$ over the vectors of V , with θ_i the angle between vector i and the nearest direction in S . Weighting by squared length keeps items with small, poorly determined effects from moving the number. The square root inverts to an angle, which is the more legible scale near the floor: an overlap of 0.19 is 64° and one of 0.0056 is 86° , nearly perpendicular rather than merely small.

Squaring rather than summing the signed dot products is not cosmetic. Because $\sum_i \alpha_i = 0$, the signed version is $\langle \sum_i \alpha_i, \beta \rangle = 0$ identically, whatever the geometry: items leaning one way along the class axis cancel those leaning the other. Squaring is also what makes the two directions of a component count equally, which is what the question requires, since projecting the class direction out costs an item the same either way.

Chance is not zero. For a randomly oriented V ,

$$\mathbb{E}[\|VS\|_F^2] = \text{tr}(V^\top V) / d \cdot r \implies \mathbb{E}[\text{overlap}] = r/d,$$

whatever V ’s rank or shape. At $d = 2048$ that is 0.0005, and Appendix A measures it directly at 0.0010. Any overlap has to be read against that number, which moves by two orders of magnitude across the widths used in this paper.

The measure is exactly invariant to per-dimension rescaling, which the standardization absorbs, and to orthogonal rotation of already-standardized data. It is **not** invariant to general linear reparameterization: we do not whiten, because whitening would make a linear representation read separable in any basis and would discard the basis the model actually computes in. Overlap values are therefore basis-relative, while the sizes and their growth with depth do not depend on the choice.

C A Construction That Is Not Same-Token

Every construction in the body keys the item on the surface form, so the token is identical in both classes of the distinction and the embedding layer has nothing to distinguish them with. That property is what licenses reading the depth profile as contextualization. Applying the same measure to a distinction that lacks the property shows what its absence looks like.

Number and tense provide that distinction. A plural or a past tense is spelled with an extra subword, so *dog* and *dog-s* are different tokens and the distinction is present in the input before any contextualization occurs.

family	model	feature	layer 0	deepest	ratio
BabyLM	opt-babylm-1.3B-20eps-seed964	Number	0.399	0.254	0.6
BabyLM	opt-babylm-125m-20eps-seed964	Number	0.344	0.221	0.6
BabyLM	opt-babylm-350m-20eps-seed964	Number	0.373	0.219	0.6
Pythia	pythia-1.4b	Number	0.197	0.124	0.6
Pythia	pythia-160m	Number	0.168	0.141	0.8
Pythia	pythia-410m	Number	0.164	0.123	0.7
BabyLM	opt-babylm-1.3B-20eps-seed964	Tense	0.391	0.152	0.4
BabyLM	opt-babylm-125m-20eps-seed964	Tense	0.305	0.137	0.4
BabyLM	opt-babylm-350m-20eps-seed964	Tense	0.360	0.138	0.4
Pythia	pythia-1.4b	Tense	0.185	0.082	0.4
Pythia	pythia-160m	Tense	0.146	0.084	0.6
Pythia	pythia-410m	Tense	0.154	0.084	0.5

Table 9: Number and tense, which are marked by an affix and therefore read at a different subword in the two classes. Unlike every construction in the body, the interaction is large at layer 0 and does not grow, which is what a distinction visible in the input looks like.

The profile inverts (Table 9). Where the same-token constructions begin near zero at layer 0 and grow by one to two orders of magnitude, number and tense begin large and grow barely or not at all. This is what a distinction carried by the input rather than built by the network looks like under this measure, and it is the pattern the body’s constructions were designed to avoid.

The lemma-keyed part of speech figures in Section 3.1 read the same way: what differs between the two keyings is the question the measure is asked, not how well it answers.

D Model Details

family	model	params	width	layers	training data	tokens
BabyLM	opt-babylm-125m	125M	768	12	BabyLM (child-available)	~2B
	opt-babylm-350m	350M	1024	24	BabyLM (child-available)	~2B
	opt-babylm-1.3B	1.3B	2048	24	BabyLM (child-available)	~2B
Pythia	pythia-160m	160M	768	12	the Pile	~300B
	pythia-410m	410M	1024	24	the Pile	~300B
	pythia-1.4b	1.4B	2048	24	the Pile	~300B

Table 10: The six models, paired by width. Training-token counts are approximate; the OPT-BabyLM models train for 20 epochs over a 100M-word corpus, so their effective token exposure is roughly twenty times the corpus size. *Layers* counts the transformer blocks, and the measure is applied to each block’s output plus the embedding layer.

The two families are matched on width rather than on parameter count, which is what the measure is sensitive to: the chance level for overlap is the class subspace rank over the width (Appendix B), so a comparison across models of different width has to be read against different chance levels. Pairing at 768, 1024 and 2048 removes that from the family comparison.

They differ in two ways that matter elsewhere in the paper. The training corpora differ by more than two orders of magnitude in size, which is the manipulation of interest. And they position tokens differently: Pythia uses rotary embeddings, applied inside attention, while OPT learns absolute position embeddings added at the input. Appendix F uses that difference to rule out absolute position as the source of the interaction.

E Deepest-Layer Measurements

Table 11 gives the deepest-layer values behind the curves of Figure 4, together with the null each overlap is read against.

distinction	item-class overlap	interaction overlap	class effect	interaction	portable fraction
POS (noun/verb)	0.0056 (5.8×, 6/6)	0.191 (5.1×, 6/6)	0.081 (29.4×, 6/6)	0.173 (6/6)	0.31 [0.25, 0.47]
role (subject/object)	0.0043 (4.5×, 6/6)	0.131 (5.5×, 6/6)	0.067 (23.7×, 6/6)	0.079 (6/6)	0.46 [0.40, 0.56]
metaphor (literal/figurative)	—	0.227 (6.1×, 6/6)	0.002 (5.5×, 6/6)	0.044 (6/6)	0.07 [0.05, 0.08]

Table 11: Every measurement the three distinctions are compared on, at the deepest layer of each model, over $n = 6$ models per distinction. The two overlaps are the two ways separability can fail by orientation: item-class overlap is the fraction of the item effect lying in the class subspace, and interaction overlap is the fraction of the interaction lying in the span of the two main effects. Each is given as the median, then in parentheses how many times its own null’s median it is and how many of the six measurements fall outside that null’s 95% interval. The null holds the effect fixed and rotates the target subspace at random 200 times, which preserves the effect’s own spectrum where a closed-form interval assuming equally weighted directions would not. The multiple is what makes overlaps comparable across models of different width, since the null itself moves by two orders of magnitude across the widths used here; the count is what says the effect holds across models rather than resting on one. The class effect and the interaction are given the same way, against their own permutation distributions: class labels shuffled within each item for the class effect, item correspondence shuffled between the two splits for the interaction. The class effect also carries a multiple, because its null is a squared length sitting on a positive floor, while the interaction’s null is a cross-split product centered on zero and only the count is meaningful. The portable fraction is the class effect’s share of the class-relevant structure, the class effect divided by the sum of the class effect and the interaction: how much of what distinguishes the two classes is a direction shared across words rather than specific to each word.

F Positional Embeddings Are Not the Source

One architectural explanation of the interaction has to be ruled out before it can be read as a fact about lexical representation. OPT-BabyLM adds a *learned absolute position embedding* at the input. The two classes of a linguistic distinction are not positionally interchangeable in a sentence, since subjects precede objects and nouns and verbs occupy different slots, so a representation that varies with position varies with the class. If that is what the decomposition is picking up, the interaction would be an artifact of where words sit rather than of how they are represented.

We re-extracted every OPT-BabyLM model with *decoder.embed_positions* zeroed and measured it identically. The ablation was verified per model rather than assumed: with the embedding zeroed, the same token at two different offsets must give identical layer-0 states, and it does, in every model (drift 0.0 to machine precision). Pythia has no such module, so there is nothing to zero there.

The right quantity is the *portable fraction* used in Table 11, the class effect’s share of class-relevant structure. If positions were manufacturing the interaction, removing them should leave a larger share of what distinguishes the two classes in a direction shared across words. Absolute sizes cannot be compared here: zeroing the embeddings in a model that was trained with them is a heavy intervention, and it shrinks every component at once, the class effect included.

model	positions	class	interaction	portable fraction
opt-babylm-125m	zeroed	0.012	0.034	0.264
opt-babylm-125m	intact	0.082	0.105	0.439
opt-babylm-350m	zeroed	0.008	0.023	0.253
opt-babylm-350m	intact	0.080	0.092	0.468
opt-babylm-1.3B	zeroed	0.009	0.035	0.198
opt-babylm-1.3B	intact	0.057	0.115	0.333

Table 12: Trained OPT-BabyLM at the deepest layer, with absolute position embeddings intact and zeroed. If positions produced the interaction, the portable fraction would rise when they are removed. It falls in all three models. Absolute sizes shrink because zeroing the embeddings damages a model trained with them, which is why the ratio rather than the level is the comparison.

Removing the position embeddings makes these representations *less* separable, not more. The portable fraction falls in every model, from 0.439 to 0.264, from 0.468 to 0.253, and from 0.333 to 0.198 (Table 12). Whatever the absolute position embedding contributes, it is contributing to the part of class

structure that *is* carried by a shared direction. Take it away and what remains is more word-specific, not less. The interaction is not an artifact of position.

A second line of evidence points the same way and needs no ablation at all. Pythia never had an absolute position embedding to remove, and its portable fractions are 0.253, 0.287 and 0.262, at or below every intact OPT-BabyLM value. The family without the mechanism shows the phenomenon at least as strongly as the family with it. An explanation resting on absolute position embeddings would have to account for the models that do not have them.

We tested this for part of speech, the construction where the positional worry is sharpest because noun and verb uses of a form occupy systematically different sentence positions. Grammatical role carries the same concern for the same reason, and behaves the same way. Metaphor does not, since literal and figurative uses of a word are not positionally distinguished.